

Program : Diploma in Polymer Technology	
Course Code : 6091A	Course Title: Plastic Products
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To identify different plastic products manufactured by various processing techniques.
- To understand the design and manufacturing of plastic products by various manufacturing processes.
- To acquire the knowledge on various finishing operations.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge about different plastics		Plastic Technology	4
Understanding about different plastic processing machinery		Polymer Processing and Machinery	4

Course Outcomes:

On completion of the course, the student will be able to

COn	Description	Duration (Hours)	Cognitive level
CO1	Outline the basic concepts of plastic product design.	13	Understanding
CO2	Explain the manufacturing of various injection moulded and blow moulded plastic products	15	Applying
CO3	Explain the manufacturing of various thermoformed and extruded plastic products.	15	Applying
CO4	Summarize miscellaneous plastic products manufacturing and finishing operations of plastic products	15	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3		3				
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the basic concepts of product design		
M1.01	List out different engineering and commodity plastic products and classify them according to their manufacturing method	3	Understanding
M1.02	State the functions of compounding ingredients used for product manufacturing	3	Understanding
M1.03	Summarize the steps involved in plastic product design	3	Understanding
M1.04	Explain the Rheological behavior of plastics and environmental impact of plastic product proliferation.	4	Understanding
Contents: Important Plastics - General idea about plastic product design. Plastic product based on commodity, engineering plastic and thermoset plastics. Classify plastic according to their manufacturing method. Functional and aesthetic design - Steps involved in plastic product design, Ingredients used in plastic products Fillers, stabilizers, curing agents, plasticizers, colorants, flow promoters, anti-degradants. Rheological behavior of plastics, stress concentration frozen in stress. Environmental impact of plastic product proliferation.			
CO2	Explain the manufacturing of various injection moulded and blow moulded plastic products		
M2.01	Describe properties of plastics used in injection molded and blow molded products manufacturing	3	Understanding
M2.02	Explain the material selection criteria for injection molding and blow molding.	4	Understanding

M2.03	Explain the manufacturing of different injection moulded plastics products.	4	Understanding
M2.04	Explain the manufacturing of different blow moulded plastics products.	4	Understanding
	Series Test – I	1	Understanding
Contents: Injection and blow moulded plastic products. Material selection criteria for injection molding and blow molding. Properties and characteristics of plastics used in injection and blow molded products- LDPE, HDPE, PVC, Nylon and Polyester. Manufacture of bucket, bottle caps, chair with formulation. Production of nylon gears and PVC chapel. Manufacture of PET bottles.			
CO3	Explain the manufacturing of various thermoformed and extruded plastic products.		
M3.01	Identify different thermoformed and extruded plastic products.	3	Remembering
M3.02	Explain the manufacturing of different thermoformed products	4	Understanding
M3.03	Explain the manufacturing process of PVC and PE pipe with formulations	4	Applying
M3.04	Explain the production of plastic film by blown film extrusion process.	4	Understanding
Contents: Thermoforming Extrusion and calendared products- various plastic products produced by Thermoforming - extrusion and calendaring. Production of disposable glass - egg trays from poly olefins. Production of PVC Pipe (flexible and rigid with typical formulation)–design factors of PVC Pipes such as SDR, Water hammer and stress State the equation and simple problems for SDR. Production of PE Pipe Line Flow diagram of PVC and PE pipe extrusion. Manufacture of plastic film - blown film extrusion.			
CO4	Summarize miscellaneous plastic products manufacturing and finishing operations of plastic products.		
M4.01	Describe the production of plastic products by casting and coating process	3	Understanding
M4.02	Explain the production of plastic products by compression and transfer moulding process	4	Understanding
M4.03	Explain rigid and flexible PU foam manufacturing	4	Understanding

M4.04	Outline various finishing and decorating operations of plastic products	4	Understanding
	Series Test – II	1	

Contents:

Miscellaneous plastic products - casting of PMMA sheets, calendaring and coating - manufacture of PVC REXIN - vinyl flooring - Typical formulations, Compression moulding - Bakelite moulded products, Manufacture of MDF and HDF. Transfer moulding - inserted plastic products, foaming - rigid and flexible PU foam - Finishing and machining of plastic products - tumbling - grinding - sanding - joining - assembling and buffing - decoration of plastic products - printing - electro plating - laser marking - vacuum metalizing.

Text / Reference:

T/R	Book Title/Author
T1	Plastic Product Design Engineering Hand book – Sydney Levy, J Harry DuBois (Springer US, 1984)
R1	Blow moulding -David A Jenners
R2	Injection moulding -Irvin Ruhin
R3	Plastic Technology Hand book - William J Patton (Publisher – Reston)
R4	Plastics Engineering Handbook Of The Society Of The Plastics Industry – M. Berins,

Online Resources:

Sl.No.	Website Link
1	https://www.hanserpublications.com
2	https://www.linkedin.com
3	apisolution.com